

Influencing factors of knowledge enhancement of corporate universities in China

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Abstract

Purpose – Corporate universities are regarded as the knowledge management institution of enterprises, but how to improve knowledge enhancement function has not been investigated. Constructed upon the knowledge-based view, the purpose of this study is to investigate the influencing factors of knowledge enhancement of corporate universities.

Design/methodology/approach – This paper takes 34 Chinese corporate universities as samples. The influencing factors and interrelationships are checked through exploratory case study and multi-case analysis. A “stimulate-action” model of influencing factors is built up based on the multi-case analysis. The fuzzy set qualitative comparative analysis is used to examine configurational paths for knowledge enhancement.

Findings – The research finding shows that only a single element of knowledge management does not constitute a necessary condition for knowledge enhancement of corporate university and explored the concurrent synergy effect and linkage matching mode of multiple conditions of knowledge management. Most of the knowledge-enhanced corporate universities are of strengthening the knowledge process, which shows that sorting out the knowledge process is the core of the corporate university.

Originality/value – Unlike the existing research, the external teachers and combing of external knowledge processes are the key factors; this paper proposed that the internal teachers and sorting out the knowledge process within the enterprise play a central role in the knowledge enhancement of corporate universities. The path of high-knowledge enhancement and without high-knowledge enhancement is asymmetry. This paper refined and extended the application of the knowledge management framework in explaining multiple conjunctural causations and improves the relevant theories of organizational knowledge management in the Chinese context.

Keywords Knowledge enhancement, Influencing factors, Configurations, FsQCA, Corporate universities

Paper type Research paper

1. Introduction

Corporate universities play an increasingly important role in the development of enterprises. According to the statistics, 80% of the Fortune Global 500 companies have established their own corporate universities (Nixon and Helms, 2002). Corporate universities can establish an efficient knowledge management system for enterprises and enhance their ability to respond quickly to changes and refine common knowledge (Chen *et al.*, 2019). The operating process of a corporate university is to build a platform to promote knowledge transfer, accumulation and innovation (Meister, 2001). Corporate universities have become the organization of corporate knowledge management, and knowledge management of corporate is the mission and core of corporate universities (Slettl, 2017; Chen *et al.*, 2019).

Nowadays, some corporate universities have problems such as unidentified enterprise knowledge gaps, unsmooth knowledge connections and weak innovation incubation capabilities, which make the knowledge innovation activities of corporate universities, have little effect. There are also some corporate universities with relatively weak knowledge service capabilities and a low professional level (Chen *et al.*, 2019). At the same time, several successful corporate universities also closely in contact with the strategic partners of the



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enterprise when carrying out the education and training of enterprise employees, optimize and integrate the training resources of the upstream and downstream industrial chains, adopt effective tools and methods to sort out the knowledge processes inside and outside the enterprise and become think tanks of the enterprise (Dealtry and Rademakers, 2005 and Scarso *et al.*, 2017). These phenomena have prompted scholars and entrepreneurs to constantly consider what factors affect the knowledge enhancement of corporate universities.

In recent years, existing research from the perspectives of human resource management and strategic management involves affecting factors of the knowledge enhancement function of corporate universities (Prince and Stewart, 2002; Dealtry, 2006 and Antonelli *et al.*, 2013). However, the existing literature seldom explores the influencing factors of knowledge enhancement of corporate university from the perspective of knowledge management (Chen *et al.*, 2019). The limited research on knowledge management in corporate universities has explored the factors of knowledge environment management, such as the configuration of hardware equipment, the latest learning technology and information-based system platforms (Sandelands, 1997; Antonelli *et al.*, 2013 and Petr and Elena, 2018). Some scholars also pay attention to the management of knowledge subjects and believe that the faculty of corporate universities is an effective carrier to support the business system (Rhéaume and Gardoni, 2016). However, the existing research has the following three limitations: First, although the existing literature involves the knowledge subject and knowledge environment of the knowledge management of corporate universities, the relevant contents of the knowledge management involved are neither comprehensive nor in-depth. More importantly, knowledge management also includes knowledge management processes (Chen *et al.*, 2019). However, the existing literature only studies the process of knowledge management from the perspective of resources and operations or knowledge transfer and has not explored the process of knowledge management in corporate universities from the perspective of combing the internal and external knowledge flow of the enterprise, resulting in incomplete research on the knowledge management elements of knowledge enhancement function of corporate universities. Therefore, it can be seen that the elements and core elements of knowledge management that promote the knowledge enhancement of corporate universities are still unclear. Second, although the existing literature has explored some activities of knowledge management to promote knowledge enhancement of corporate university, it does not have a framework of consistent influencing factors based on theoretical construction. Third, existing researches often limited to exploring the net effect of a single influencing factor while ignoring the combined effect of multiple factors. Undoubtedly, the knowledge management of a corporate university as a complex knowledge activity is the result of multiple factors from different levels and different subjects. The kind of causal complexity problem cannot fully explain the causal complexity and the synergistic effect among multiple factors by exploring the net effect of a single factor or the moderating effect of at most three variables through traditional regression analysis (Fiss, 2007 and Misangyi *et al.*, 2017). For the above reasons, it is still necessary to answer the question of how to improve the knowledge enhancement of corporate universities from the perspective of holistic.

At present, research on the function of knowledge management undertaken by corporate universities in a specific environment is gradually emerging. The existing literature has studied the knowledge management of corporate universities in Germany (Andresen and Lichtenberger, 2007), Russia (Slettl, 2017), Britain (Prince and Allison, 2003) and Italy (Guerici *et al.*, 2010), but there is no literature on the knowledge enhancement of corporate universities in China.

From this perspective, the following major research questions are raised:

RQ1. What are the factors that influence the knowledge enhancement of corporate universities?

RQ2. How can the combination and interaction of knowledge management elements of corporate universities affect the knowledge enhancement of corporate universities?

RQ3. What are the core elements of knowledge management that promote the knowledge enhancement function of corporate universities?

The paper is organized as follows: First, this study reviewed the research results in the field of corporate university knowledge management, explored the elements of corporate university knowledge management through exploratory case study and multi-case analysis and sorted out the impact of each element on the knowledge enhancement of corporate universities. Next, this study adopts the method of fuzzy-set qualitative comparative analysis to explore which combination of knowledge management elements can more efficiently promote the knowledge enhancement of corporate universities and which combination of knowledge management elements inhibit the knowledge enhancement of corporate universities. Based on the results of statistical analysis, this paper further theoretically summarizes the combination of knowledge management elements of the corporate university and puts forward theoretical propositions that can be tested by subsequent research to promote the knowledge enhancement of corporate university.

2. Literature review

2.1 Knowledge management

Knowledge management is not only the management of knowledge but also the management of various tangible and intangible resources related to knowledge (Dalkir, 2005; Gao *et al.*, 2008; Denford and Ferriss, 2018). From the perspective of the subject of knowledge management, some scholars believe that knowledge management is mainly the management of knowledge employees or knowledge assets (Alavi and Leidner, 2001; Gold *et al.*, 2001 and Prof *et al.*, 2014). Some scholars also believe that creating an environment for enterprise knowledge management is the basis for implementing knowledge management (Schneckenberg *et al.*, 2015). The environment of knowledge management involves enterprise information technology infrastructure, organizational design and organizational culture (Bhatt, 2001). Some scholars have explored the process of knowledge management and believe that knowledge management is a systematic process from the perspective of the knowledge flow (Bamel and Bamel, 2018). As Prof *et al.* (2014) proposed, knowledge management refers to the process of creating, acquiring and using knowledge to enhance the performance of an organization. Denford and Ferriss (2018) believe that knowledge management should pay attention not only to the collection, classification, storage, query and reuse of knowledge itself but also to be closely integrated with the core business processes of the enterprise. Process-oriented knowledge management aims to integrate business processes and knowledge management (Khuram *et al.*, 2016).

Most of the existing research examines the basic elements of the knowledge management support system from three aspects: the means of knowledge activities, the management of knowledge processes and the management of knowledge subjects (Dalkir, 2005; Prof *et al.*, 2014; Schneckenberg *et al.*, 2015; and Denford and Ferriss, 2018). The completeness of the elements of the knowledge management system is a necessary condition for it to effectively support and ensure the operation of knowledge management (Bamel and Bamel, 2018). Knowledge management is precisely through a combination of the three types of knowledge management elements to achieve the sharing and integration of knowledge within an organization (Alavi and Leidner, 2001).

The existing research focuses on the supporting elements of enterprise knowledge management (Scarso *et al.*, 2017 and Chen *et al.*, 2019), while there are few studies on the knowledge management elements of the corporate university. Corporate universities are open cross-border organizations, and their knowledge management standpoints and methods are bound to be different from those of enterprises and universities. Few scholars

have conducted in-depth research on the knowledge management of corporate universities. It is necessary to further study the particularity of the knowledge management of such cross-border organizations. The few existing documents mainly explore the composition of the factors that influence the knowledge function of the corporate university from the aspects of constructing the organizational structure of the corporate university, the construction of the information system, the construction of the lecturer team and the course development (Doody, 2001; Clinton *et al.*, 2009 and Tsipes *et al.*, 2016). Some scholars also believe that enterprises can enhance the knowledge of corporate university by arranging manpower, material resources, financial resources, management systems and process specifications (Akram, 2002; Guerri *et al.*, 2010 and Baporikar, 2015). Although some documents involve the subject of knowledge management, the environment of knowledge management and the process of knowledge management, the specific dimensions are not clear or comprehensive. The knowledge management of corporate universities is the key to improving organizational capabilities (Chen *et al.*, 2019). However, most studies explore the process of knowledge transfer in corporate universities and have not yet explored how to sort out the knowledge process of enterprises (Sandelands, 1997; Paton *et al.*, 2005; Schultz, 2015). To sum up, the existing literature has not yet conducted in-depth and comprehensive research on knowledge management elements of corporate universities, such as the subject of knowledge, the process of knowledge activities and the knowledge activities environment.

Besides, the existing research is limited to a single perspective of an element of knowledge management, and it is not clear how multiple elements cooperate and interact to affect the function of corporate universities, which ignores the matching relationship and coordination mechanism between knowledge management combination and knowledge enhancement. In fact, the knowledge management activities of corporate universities are a complicated process, and it is urgent to explore the phenomenon of knowledge management from a diversified and integrated perspective. Given this, it is necessary to study the multiple concurrent factors and causal complex mechanisms that affect the enhancement of knowledge of the corporate university. It is also necessary to further study how to link and match the multiple antecedent conditions that affect the enhancement of knowledge of the corporate university and how to maximize the effect of knowledge management.

2.2 Knowledge enhancement functions of a corporate university

Meister (1997) first defined a corporate university as an educational and training institution to improve the performance of employees. All previous studies agreed on this view, holding that corporate universities are the extension and expansion of traditional education. Since universities cannot meet the needs of enterprise knowledge and industry knowledge required by enterprises, the corporate universities should be established to undertake the continuing education of working adults (Nixon and Helms, 2002 and Akram, 2002). With the further development of corporate universities, an increasing number of scholars begin to realize that corporate universities can help enterprises to obtain sustainable core competitiveness in the increasingly fierce business environment around their strategies and development goals. Most corporates have endowed corporate universities with more and more complex strategic functions (Paton *et al.*, 2005 and Antonelli *et al.*, 2013).

Many scholars pointed out that corporate universities are strategic tools designed around enterprise strategies and development goals to enhance enterprise competitiveness. Nixon and Helms (2002) believe that the main function of corporate universities is to undertake the function of knowledge transfer, and to improve the ability and productivity of employees through education and training. With the continuous development of corporate universities, the training objects of corporate universities are no longer limited to employees of the enterprise, but have been extended to employees of upstream and downstream industry

chain (Rheume and Gardoni, 2016). Scarso *et al.* (2017) are concerned that the corporate university is an integrator of internal and external knowledge resources, and has the capability of “bridge” for knowledge dissemination and diffusion between internal and external enterprises. Merister (2001) pointed out that corporate university is the post-university continuing education and also an integral part of lifelong education. Baporikar (2015) emphasized that the corporate university has established a knowledge sharing platform, which is an exchange center for internal and external knowledge circulation and improves the efficiency of knowledge transfer.

In recent years, corporate universities have not only played a role in the construction of internal and external communication platforms, but also promoted knowledge innovation (Patrucco *et al.*, 2017; Sletli, 2017). Scarso *et al.* (2017) pay attention to the important role played by corporate universities in promoting the development and maturation of knowledge, and believe that corporate universities are the knowledge centers of enterprises. Patrucco *et al.* (2017) emphasized that the capability of corporate universities includes not only knowledge transfer or dissemination, but also knowledge development and creation. Some scholars even believe that corporate universities have integrated and nurtured new knowledge, and promoted the innovation activities of enterprises through industry-university-research cooperation (Rheume and Gardoni, 2016; Chen *et al.*, 2019).

According to the existing literature, we find that in a certain sense, the knowledge management capabilities of mature corporate universities are more systematic and comprehensive, that is, the function system of corporate universities not only includes the organic connection between traditional formal education and lifelong education and further upgrading education and other training activities, but also should actively participate in knowledge incubation.

3. Research method and sample selection

3.1 Research method

To identify the key influencing factors and to explore how these influencing factors affect the knowledge enhancement of corporate university in the Chinese context, the case study and the survey are both appropriate research methods. Yin (2003) pointed out that multiple case studies can repeatedly verify the same logic, which is similar to doing multiple experiments and is conducive to objectively and comprehensively reflecting the facts. Therefore, the multi-case study is applicable to this paper. The multi-case study mainly consists of two parts. The first part is the identification of the key influencing factors of knowledge enhancement and detection of the relationship among these factors by exploring the two sample cases of corporate universities. Through exploratory research, several propositions are put forward to illustrate how corporate universities can enhance their knowledge enhancement functions. The second part is the empirical research, which verifies and tests the proposition brought forward in the first part through multi-case analysis of 34 sample cases. The 34 typical sample corporate university were analyzed by selected experts and the knowledge enhancement effect of each sample corporate university is judged according to identified influencing factors, and the cross-case influencing factor matrix is build up to verify and explain the propositions brought forward in the first part. Based on above two parts of study, a theoretical framework is built up to show how corporate universities can achieve knowledge enhancement.

Then fsQCA is applied to compute causal configurations for elucidating the multifaceted conditions (i.e. combination of independent variables, which leads to the specific outcome) (Elena *et al.*, 2018). The reason for adopting this research method is that it can explore the complex interplay between many causal factors based on a combined perspective. It is suitable for exploring the combination of factors rather than the impact of individual factors on the results (Douglas *et al.*, 2020). In order to reveal the influence mechanism of knowledge

enhancement in corporate universities in China, it is far from enough to use conventional statistical analysis to analyze the independent or dual interaction of various knowledge management, and it is necessary to explore the complex interaction of various factors from the holistic perspective (Chen *et al.*, 2019). Precisely, fsQCA investigation enriches the multi-case study outcomes by identifying the grouping of influencing factors, KM processes for increasing knowledge enhancement (Slettli, 2017). Misangyi and Acharya (2014) support that fsQCA is an innovative technique to identify different associations for management situations and difficulties.

3.2 Sample selection

This paper selects 34 corporate universities affiliated with technical and skilled enterprises that have been established for more than three years as case studies (Table 1). The conditions that the research samples need to meet and the basis for screening are as follows: First, the principle of selection typicality (Eisenhardt and Graebner, 2007). The corporate university founded by the enterprise in the technical field is based on the rapid rate of technology update of this type of enterprise, which puts forward higher requirements for the knowledge management of the enterprise. Therefore, the knowledge activity services undertaken by the corporate universities established by such enterprises are also more diverse. Second, the principle of data availability (Yin, 2003). The public information of these 34 corporate universities are relatively rich, and researchers have the opportunity to directly understand the relevant information and obtain first-hand information through field research. In these 34 typical corporate universities, two samples are for the exploratory study bringing forward the propositions about influencing factors and relationship identification for theory building; another 32 samples are for the verification, confirmation of the propositions, providing empirical evidences. The research cases and variables are relatively stable within 3 years. In the analysis of fsQCA, there are positive cases with high knowledge enhancement function and negative cases with non-high knowledge enhancement function, which is conducive to exploring the conclusion of generalization. Table 1 shows the list of the 34 selected sample corporate university and their affiliated companies.

3.3 Data collection

This paper mainly adopts two different data collection methods: First, In-depth interviews. The authors of this paper conducted field surveys of 34 sample corporate universities from May 2019 to December 2020. Each interview lasted about 2–3 hours. The number of samples collected through interviews and investigations in this paper has reached more than 90%. The field research mainly includes three aspects: industry expert consultation, participation in industry conferences, participation in well-known corporate university forums, and relationship networks (kinship, friend relationship and enterprise relationship, etc.) to interview and obtain corresponding data from a number of corporate university executives (who must have served for more than 3 years), students from corporate universities, enterprise executives and personnel working in senior professional manager training institutions. In the process of investigation, the initial interview designed an open interview outline, mainly in a qualitative way. The follow-up interview designed a semi-open outline including the questionnaire items, and gradually involving quantitative issues such as variable coding and assignment. Using an anonymous survey method, 242 questionnaires were distributed in 34 corporate universities, including 223 were valid questionnaires, and the effective questionnaire recovery rate was 92%. Second, second-hand materials. The authors and members of the research team accumulate second-hand materials through internal information such as the work summary and memorabilia of the corporate university, websites, news reports, principal interviews with the media and related academic articles, so as to fully understand the relevant data of the corporate university. In order to ensure the

No	Name of corporate university	Owned enterprise	No	Name of corporate university	Owned enterprise
1	ZTE Institute	ZTE Corporation	18	Huali School of Management	Holley Group
2	Huawei University	HUAWEI Corporation	19	Dongling College	Dongling Kaiqin Group
3	Goldwind University	Goldwind Technology Company	20	Far Eastern University	Far East Holding Group
4	Baosteel Talent Development Institute	Baosteel Group	21	Delta Enterprise College	Delta Group
5	Trinity College	Sany Heavy Industry Group	22	CRRC University	CRRC Group
6	China Resources University	China Resources Group	23	TCL Leadership Development College	TCL Group
7	Changan Automobile University	Changan Enterprise Group	24	UFIDA College	UFIDA Group
8	GCL University	GCL Group	25	China General Nuclear Power University	China General Nuclear Power Group
9	Shenhua School of Management	Shenhua Group	26	East University	East Group
10	Mindray School of Management	Mindray Medical Company	27	Datang University	Datang Telecom Technology Industry Group
11	Haier University	Haier Group	28	Lenovo College	Lenovo Group
12	Kingdee University	Kingdee International Software Group	29	Flextronics College	Flextronics Group
13	Xinao University	Xinao Group	30	Hisense College	Hisense Group
14	China Telecom Institute	China Telecom Group	31	BOE University	BOE Technology Group
15	AVIC University	China National Aviation Corporation	32	HP University of China	China Hewlett–Packard
16	Konka College	Konka Group	33	Skyworth College	Skyworth Group
17	Midea College	Midea Group	34	State Grid Institute of Technology	State Grid Corporation

Source(s): Official websites of corporate universities in China

Table 1.
The profile of the
sample corporate
universities

reliability and validity of the study, this study follows principles of the triangulation method put forward by Yin (2003) in data collection and data analysis aspects, with many complementary sources of evidence.

3.4 The case analysis based on two typical cases for exploratory study

The two selected sample cases for exploratory study in the first part are as followed. The first one is China Resources University (CRU). The second one is the Huawei University (HU). There are two reasons for selecting them as samples for exploratory study. Firstly, these two cases are very typical, and both of them made good performances in knowledge enhancement. Secondly, two investigators in our investigation group just come from above two CU, providing available and sufficient primary data with high-quality.

3.4.1 The case sketch of China resources university. In 2012, China Resources University was established, and it is jointly responsible for the training of employees with its nine professional

colleges. In 2014, CRU established a platform for connecting internal and external knowledge of the enterprise, focusing on the internal sorting and external dissemination of knowledge. In addition, CRU has also cooperated with famous universities, research institutes and consulting companies at home and abroad to further expand the scope of teacher selection and extend the training target to China Resources' upstream and downstream industrial chain enterprises. In recent years, CRU has paid more and more attention to innovative products and entrepreneurial incubation, and has established an open innovation platform, innovation laboratory and expert database to provide students with services such as integration of intelligence, financing and technology.

3.4.2 The case sketch of Huawei University. Huawei University was established in 2005, focusing on employee induction, business training and mentor guidance, and providing online real-time training for employees in more than 40 countries by using distance education and virtual classrooms. HU established a perfect qualification system for employees' skills and abilities, closely combining training with business, and supporting the implementation of enterprise strategy and the appreciation of human capital through empowerment. Focusing on the improvement of project management capabilities and professional capabilities, HU has developed more than 680 hours of courses in cooperation with business departments. In terms of external relations, HU conducts in-depth cooperation with relevant institutions inside and outside the enterprise to promote knowledge integration of the industry chain by introducing and internalizing cutting-edge knowledge of the industry and customer demand information. As a contact unit, HU has established more than 20 innovation centers, 26 capability centers and laboratories around the world. These institutions also focus on cutting-edge knowledge exploration and innovation project incubation.

4. The exploratory findings of knowledge enhancement from CRU and HU

4.1 Internal teachers of the corporate university (IT)

The internal teachers are the full-time and part-time teachers of corporate universities that employ internal technical experts, excellent employees, department managers and other front-line employees (Sharma and Pandher, 2018). Teachers in corporate have rich enterprise knowledge and practical experience. They can provide training and practical guidance to trainees, and can more specifically solve professional technical problems (Guerci *et al.*, 2010). The internal teachers of a corporate university are often composed of middle and senior executives or senior engineering and technical personnel of the enterprise, which has strong enterprise characteristics when implementing the unique strategic plan of the enterprise (Chen *et al.*, 2019). However, if the contents taught by senior executives and senior technicians cannot be well connected and coordinated with each other, and there is some deviation in orientation, the strategy of the enterprise cannot be well implemented.

An interviewee at CRU said:

CRU has more than 500 full-time managers as teachers, 3,800 managers from key business positions within the Group. Most of CRU's internal teachers select technical experts, outstanding employees, department managers and other front-line employees according to the needs of business departments.

One of the interviewees at HU said:

HU's chairman and other executives usually also participate in teaching trainees. Internal teachers not only undertake the function of teaching and training for employees, but also carry out corresponding research and provide solutions around various problems of the company, playing the role of a think tank within the enterprise.

P1. For corporate universities, internal teachers of the corporate university contribute significantly to knowledge enhancing.

4.2 Teachers employed outside the corporate university (TEO)

External teachers are full-time and part-time lecturers employed by enterprises from outside to teach employees of enterprises (Mors, 2010). Most of the external teachers are well versed in cutting-edge theoretical knowledge, have broad industry knowledge and rich experience in enterprise training, but they are not clear about the operation of the enterprises affiliated to the corporate university and the strategic intentions of the top managers, and the knowledge they teach generally lacks practical and pertinence, which cannot be effectively matched with the actual business of the students (Meister, 2001). Therefore, how to select external teachers suitable for enterprises is also an important support for corporate universities to carry out training and knowledge innovation (Rhe ume and Gardoni, 2016). However, the familiarity of high-quality external teachers with the enterprise should be the most important criterion to judge whether the external teachers meet the requirements of the corporate university (Duysters and Lokshin, 2011; Chen *et al.*, 2019).

As noted by an interviewee from CRU:

CRU has more than 1,000 part-time tutors. Innovation or entrepreneurial projects require internal teachers who are familiar with corporate processes to train and guide in practice. At the same time, professors and scientists from universities who understand the latest cutting-edge industrial technology information are employed as external teachers to provide intellectual support for innovation and exploration. However, because many external teachers are not familiar with the company's processes, CRU will organize external teachers to conduct research on the company, so that they are familiar with the group's business processes.

Another HU interviewee also noted:

HU has hired high-paying research teams from universities and research institutes to make up for its lack of teachers. Some external teachers are familiar with the chairman or principal, and these teachers have a better understanding of the management philosophy of the group's executives.

P2. For corporate universities, teachers employed outside the corporate university contributes significantly to knowledge enhancing.

4.3 Sorting out the knowledge process within the enterprise (SKPW)

Internal knowledge process combing is to use the specific methods and experience of knowledge management to deal with the internal business processes of the enterprise, and to establish corresponding methods and mechanisms to store and accumulate the knowledge generated by various stages and departments within the enterprise (Martin, 2015). A corporate university is an organization established by an enterprise to carry out knowledge management for the enterprise (Scarso *et al.*, 2017; Chen *et al.*, 2019). However, the concept of knowledge management has long been integrated with the business process of the enterprise (Patrucco *et al.*, 2017). On the one hand, corporate universities explore the parts and contents of the business department of the enterprise that is conducive to or hinder knowledge transfer, and finds the internal logic between them (Rhe ume and Gardoni, 2016). On the other hand, corporate universities explore the knowledge and skills of the business department and distribute them to the department with the greatest output (Patrucco *et al.*, 2017).

An interviewee at CRU said:

CRU investigates whether the relevant professional knowledge of the internal knowledge department of the enterprise is missing, and judges the mastery of the engineering science knowledge and technical science knowledge of the business department. At the same time, CRU identifies knowledge and resources required by business departments, tracks internal knowledge

weaknesses and dynamic changes of industry knowledge, and helps business departments such as production, R&D, and marketing departments to obtain knowledge that matches their work.

A HU interviewee noted:

HU provides a series of learning tools and methods, embedding the new concept of adapting to change, so that students can integrate into actual business operations after the exercises in class. Since HU has been accumulating and innovating knowledge inside the enterprise rather than outside the enterprise, HU pays more attention to sorting out and studying the internal knowledge process of the enterprise. For example, it provides information and technical consulting services for R&D departments, and conducts production-oriented basic research.

A HU interviewee expressed a similar view:

Each business unit brought the department's task objectives and weaknesses in the business process to HU. The key members of the business department formed a problem-solving team to participate in the project. Through action learning, the professional performance improvement teachers of HU guide to solving the knowledge weaknesses of the business department.

P3. For corporate universities, sorting out the knowledge process within the enterprise contributes significantly to knowledge enhancing.

4.4 Sorting out the external knowledge process of enterprises (SEKP)

The external knowledge process is to search for knowledge and information of the external knowledge network, and introduce the independent and scattered related skills and knowledge information such as market information, customers, products and process knowledge required by the enterprise into the internal knowledge base of the enterprise (Brunswick and Vanhaverbeke, 2014). Knowledge is further internalized to improve the knowledge management and innovation capabilities of the enterprise. Corporate universities are cross-border and open organizations. Strengthening knowledge exchange with external organizations can build a complementary knowledge association linked by knowledge. The development of enterprise products requires the use of cross-border organizations to absorb mature technologies from outside and acquire research and development resources to cultivate new technologies (Patrucco *et al.*, 2017; Slettli, 2017). As an intermediary, the corporate university has set up an interface for connecting internal and external knowledge, using multi-channel innovative resources outside the enterprise to obtain knowledge production resources, and providing a suitable knowledge source for the effective operation of the enterprise's knowledge process through reasonable allocation of knowledge resources (Chen *et al.*, 2019).

A CRU interviewee noted:

CRU has built a bridge between enterprises and famous universities and scientific research institutions, and introduced new technologies and new ideas to promote the transformation and application of scientific and technological achievements. For example, for China Resources' power generation conversion gas differential pressure project, CRU explored that Tsinghua University had purchased similar technologies from the United States, so it contacted Tsinghua University's international technology transfer platform and put the natural gas pressure energy recovery power generation project of China Resources Gas Pipeline into operation with its rotor expander technology.

Another interviewee at HU also noted that:

In 2019, HU and Huazhong University jointly trained a master of engineering, and organized benchmarking enterprise project management exchange meetings and other activities to explore and integrate external knowledge resources, thus improving the innovation ability of enterprise personnel. HU has also introduced the concept and tools of performance improvement and adopted training strategies to solve weakness problems and cultivate employees' capabilities simultaneously.

- P4. For corporate universities, sorting out the external knowledge process of enterprises contributes significantly to knowledge enhancing.

4.5 Technical means (TM)

The application of technical means is one of the important factors for effective knowledge management in corporate universities (Cerchione and Esposito, 2017; Guerzi *et al.*, 2010). Corporate universities are not only equipped with basic facilities, including the construction of campuses and teacher seminar rooms, but also equipped with corresponding teaching technology facilities in these areas, such as multimedia equipment and teaching supporting technology. In the internet era, advanced modern technology has also brought opportunities to corporate universities. More and more corporate universities have begun to use information technology to better carry out knowledge management (Alagaraja and Li, 2015).

An interviewee from CRU explained:

Based on the concept of creation and sharing, it has developed an APP learning platform of "Creation and communication". The shared learning method has further accumulated and promoted knowledge, making CRU a think tank for enterprise development.

Similarly, a HU interviewee said:

HU uses Internet learning resources, integrates E-learning and other technologies to build a mobile network education platform and online information database, which has become an important development strategy for corporate universities.

- P5. For corporate universities, technical means contributes significantly to knowledge enhancing.

4.6 Knowledge sharing platform (KSP)

Compared with traditional learning institutions, corporate universities carry out a kind of workplace learning, that is, "work is learning" (Patrucco *et al.*, 2017). The knowledge sharing platform created by the corporate university promotes the knowledge exchange and interaction between employees within the enterprise, and further accumulates and enriches the diverse knowledge among organizations. The corporate university is a knowledge sharing platform that realizes the interactive transformation, knowledge exchange, sharing and utilization of tacit knowledge and explicit knowledge inside and outside the enterprise (Baporikar, 2015). The essence of knowledge sharing is mutual communication and exchange (Nonaka *et al.*, 2000; Yang *et al.*, 2019; Obermayer and Toth, 2019). Corporate universities create an atmosphere of integrity and openness, especially in the teamwork and continuous learning of the project to enable trainees to cooperate and help each other (Guerzi *et al.*, 2010; Abel and Li, 2012). Trainees are more active in exchanging and sharing experiences and knowledge, and will not deliberately reserve them for fear of harming their interests. The harmonious relationship between teachers and the smoothness of education and guidance activities are also conducive to enhancing mutual trust and improving the team's innovative ability.

One of the interviewees at CRU said:

CRU aggregate knowledge resources, human resources, and technical resources to provide a corresponding environment and atmosphere for knowledge sharing and creation. CRU not only builds a hardware platform for knowledge sharing and application for organizational learning, but also promotes communication and exchange inside and outside the enterprise, especially creating a mutual trusting psychological environment, and improving the exchange and sharing of explicit and tacit knowledge.

An interviewee at HU said:

HU provides a good atmosphere for trainees to discuss and communicate. Through the exchanges and learning of HU, misunderstandings and differences are eliminated, and employees are more motivated to build consensus and engage in knowledge creation activities.

Another interviewee at HU said:

HU brought gathered employees from different departments to form a case study group of the project. After training and teamwork, it formed a common understanding and created an open and innovative atmosphere for knowledge sharing and creation.

P6. For corporate universities, knowledge sharing platform contributes significantly to knowledge enhancing.

Teachers in corporate universities have proprietary knowledge. Teachers transfer individual knowledge into group knowledge and then group knowledge into organizational knowledge. Teachers are the basic resources of corporate universities, which directly affect the training work of corporate universities (Meister, 1997). The teachers of corporate universities are composed of internal full-time and part-time teachers and external part-time teachers (Dealtry, 2006). Subjects of knowledge activities include individuals, groups, and organizations (Denford and Ferriss, 2018). Corporate universities are cross-border and open organizations. Strengthening knowledge exchange with internal and external organizations can build the complementary knowledge association linked by knowledge. The process of knowledge activities refers to a series of specific activities such as knowledge acquisition, arrangement, transfer, sharing, application and innovation inside and outside the enterprise (Alavi and Leidner, 2001). Khuram *et al.* (2016) proposed that the process management of knowledge will affect the production effect of enterprises. Thus, sorting out the knowledge process within the enterprise and sorting out the external knowledge process of enterprises can be considered as the process of knowledge activity. Corporate universities have built a hardware platform for knowledge sharing and application for organizational learning (Schneckenberg *et al.*, 2015; Patrucco *et al.*, 2017). At the same time, corporate universities aggregate knowledge resources, human resources, and technical resources to provide a corresponding environment and atmosphere for knowledge sharing and creation. The environment of knowledge activities refers to the circumstances and conditions for knowledge transfer and knowledge creation (Darroch, 2005). The environment of knowledge activities is a shared space that promotes knowledge exchange and creates relevance (Bhatt, 2001). Technical means and knowledge sharing platform can be regarded as knowledge activity environment. According to the exploratory study and empirical evidence, an “Subject-Process-Environment” model of knowledge enhancement of corporate university is built up based on the six propositions put forward in part 1 (see Figure 1).

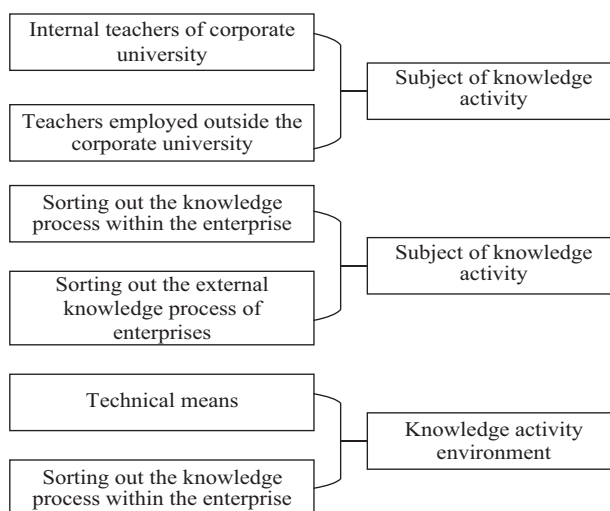


Figure 1.
“Subject-Process-
Environment” Model of
knowledge
enhancement of
corporate university

5. Multi-case analysis for empirical evidence

The investigation group selected the other 32 typical sample cases for verification and confirmation, which may increase the validity of this paper and improve the persuasiveness of the research findings.

5.1 The key influencing factor verification based on the qualitative data analysis

The software of Atlas.ti 8 was applied to process the qualitative data analysis, exploring the influencing factors of knowledge enhancement of corporate university in the Chinese context. Referring to the data analysis procedures put forward by [Miles and Huberman \(1994\)](#). First, the collected original qualitative datas about knowledge enhancement were checked and imported into Atlas.ti 8. The key words coding was carried out based on the full comprehension of the qualitative data, and 17 variables were induced, including the level of teachers in the enterprise, the degree of cross-interface knowledge convergence, the communication atmosphere created and so on (see in [Table 2](#)). Second, according to the relationships among the 17 variables, key core factors were sorted out and summed up. The six key core factors correspond to the propositions from exploratory study in part 1, including internal teachers of the corporate university, teachers employed outside the corporate university, sorting out the knowledge process within the enterprise, sorting out the external knowledge process of enterprises, technical means and knowledge sharing platform. The empirical evidence from part 2 is consistent with the exploratory study findings in part 1. At last, all the coding and quotation were handed to other two experts and other four academic researchers to examine and verify. The reliability and validity passed the examination, and the examining process was omitted due to the limitation of paper length.

5.2 The relationships among influencing factors based on multi-case analysis

After the identification of the influencing factors, the relationships among these factors and corresponding mechanism were explored and clarified by the 2 experts and 4 academic researchers in the investigation group. According to the face to face communication between the investigation group and executives of the 32 corporate universities, the agreements were achieved and the data in investigation table were finally acquired shown in [Table 3](#). Because

Table 2.
Key influencing factors
and variables of
knowledge
enhancement of
corporate university

Key word coding of the case data	Identified variables	Key factors
Part-time lecturer Engineers as teachers Department managers as teachers Full-time lecturer Senior management participate Interactive Supplementary teaching content *Same idea Communicate**corporate strategy	The level of teachers in the enterprise	Internal teachers of corporate university
Content* Mutual acceptance conform*development strategy * Indoctrination Executive support Familiar with**strategic planning Familiar with**technology Familiar with**business process	The correlation degree of teaching contents among senior management teachers The support degree of teaching content between senior managers and middle managers	Teachers employed outside the corporate university
Relationship with*executives Familiar with executives*management Familiar with*the ideas of executives	Familiarity of external teachers with enterprise knowledge process The familiarity of external teachers with the senior management of the enterprise	
External teachers*ratio Hire*experts Senior executives* hired personally Principal personally **selected and hired	Number of external teachers employed by senior executives	
Requesting research Analyze the existing*problems Analysis*business barriers Identify*shortcomings Job ability analysis Understand*knowledge mastery	Extent of identifying knowledge gaps	Sorting out the knowledge process within the enterprise
Tracking new technologies Familiar with*business process Prepare*business process documentation Refine*knowledge Frontier knowledge Track industry trends	Explore the level that matches the knowledge required by the enterprise	
Cross-departmental*collaboration Project**cooperation Communication between departments Joint Conference System Attend high-level meetings case sharing *information exchange	The degree of cross-interface knowledge convergence	
Collect market information Search*external knowledge Identify *external knowledge Diagnose the pros and cons	Level of enterprise external knowledge assessment	Sorting out the external knowledge process of enterprises
Host forum Hold a meeting Build*link platform Technology exchange Customer relationship management Value chain integration R&D alliance matching**network market conjunction	Closeness of connection with enterprise external knowledge sources	
Introduce*intellectual resources Introduce*technical resources Introduce quality courses Introduce advanced management methods Establish an expert database	Degree of introduction of enterprise external knowledge	
Mobile network Perfect* multimedia equipment E-learning Online learning resources Online forum	The perfection degree of information network platform	Technical means
Database construction Learning management system Case library*construction Industry Information*Database MOOC courses Compile books	Level of database and document coding	

(continued)

Table 2.

Key word coding of the case data	Identified variables	Key factors
*Discussion WeChat group communication Participate in the exchange meeting *teamwork Mutual assistance at work knowledge complementarity Harmonious student relationship communication mechanism *learning atmosphere Reading salon Hold a case contest *Psychological environment mutual-trust Teacher-student communication Smooth teaching activities class interaction Harmonious teacher-student relationship	Extent of interaction between participants The communication atmosphere created Teacher-student relationship and degree of communication	Knowledge sharing platform
Note(s): *Shows wildcard; shows Boolean logic		

of the length of the paper, the survey results of 22 corporate universities was omitted in Table 3. To detect how the above identified influencing factors affect the knowledge enhancement and in what degree they affect, the function of knowledge enhancement was observed and estimated. The investigation group carefully considered both the connotation of knowledge enhancement and characteristics of corporate universities, and decided to take four indicators to display the function of knowledge enhancement, including cohesion education function, knowledge upgrading function, knowledge connection function and knowledge incubation function. Based on the field study and analysis of cross-cases matrix of influencing factors, the investigation group conducted several rounds of meetings to explore the role of the factors and the relationships among them. In the end, the empirical findings supported that the propositions in exploratory study.

The corporate university is the knowledge management organization of the enterprise, and supporting the knowledge management within the enterprise to which it belongs in its original intention (Baporikar, 2015; Sletli, 2017; Scarso *et al.*, 2017). Only by investigating the lack of knowledge related to the enterprise, can corporate universities further clarify what kind of matching knowledge resources need to be sought for the various departments and employees of the enterprise, which experts to hire and what content he teaches, and how to configure the teaching infrastructure used. Sorting out the internal knowledge process of corporate universities plays a central role in the elements of knowledge management. At the same time, the knowledge enhancement of the corporate university will inevitably be accompanied by the continuous strengthening of internal teachers. Through the configuration analysis of the subject of knowledge management, the process of knowledge management and the environment of knowledge management in each stage of corporate university, it is shown that when the knowledge management capability of corporate universities develops to a high-level stage, the internal teacher is often the key factor for knowledge enhancement. Most of the external teachers in the corporate university teach theoretical knowledge, while the internal teachers are all enterprise executives or technical experts with practical experience. Compared with external teachers, the internal teachers are more in line with the business practice of teaching enterprises due to their actual practical experience. Therefore, the actual operation and business skills of the business department of the enterprise still need to rely on internal teachers to impart experiences and methods. Besides, the managers and senior executives inside the enterprise as teachers can better implement the strategic plan of the enterprise. At the same time, it is found that internal teachers of the corporate university, teachers employed outside the corporate university as knowledge subjects, providing the guarantee for the knowledge enhancement of corporate

Table 3.
Cross-cases matrix of
influencing factors

Sample CU number factors		1	2	3	4	5	6	7	8	9
Internal teachers of corporate university	The level of teachers in the enterprise (strong or weak)	*****	****	*****	*****	****	****	****	**	**
	The correlation degree of teaching contents among senior management teachers(whether or not)	*****	****	****	*****	*****	*****	**	**	**
	The support degree of teaching content between senior managers and middle managers(strong or weak)	*****	****	****	*****	*****	****	**	**	**
Teachers employed outside the corporate university	Comprehensive evaluation	++++++	+++++	+++++	+++++	+++++	+++++	++	++	++
	Familiarity of external teachers with enterprise knowledge process(strong or weak)	****	*****	*****	*****	*****	*****	++	++	++
	The familiarity of external teachers with the senior management (strong or weak)	****	*****	****	*****	*****	****	**	**	**
Sorting out the knowledge process within the enterprise	Number of external teachers employed by senior executives(strong or weak)	****	*****	*****	***	*****	****	**	**	**
	Comprehensive evaluation	+++++	+++++	+++++	+++++	+++++	+++++	++	++	++
	Extent of identifying knowledge gaps(strong or weak)	*****	*****	*****	*****	*****	**	0	0	0
	Explore the level that matches the knowledge required by the enterprise(strong or weak)	*****	***	****	*****	**	****	0	0	0
	The degree of cross-interface knowledge convergence(strong or weak)	*****	*****	*****	*****	*****	**	**	**	**
Comprehensive evaluation		++++++	+++++	+++++	+++++	+++++	++	+	+	+
(continued)										

Table 3.

Sample CU number factors		1	2	3	4	5	6	7	8	9
Sorting out the external knowledge process of enterprises	Level of enterprise external knowledge assessment(strong or weak)	*****	***	****	*****	**	**	**		
	Closeness of connection with enterprise external knowledge sources(strong or weak)	*****	****	****	***	***	**	**	0	**
	Degree of introduction of enterprise external knowledge(strong or weak)	*****	*****	*****	*****	***	**	**	**	0
Technical means	Comprehensive evaluation	+++++	++++	+++++	+++++	++++	++++	++++	++++	+
	The perfection degree of information network platform(strong or weak)	*****	*****	*****	*****	*****	*****	*****	*****	*****
	Level of database and document coding(strong or weak)	+++++	+++++	+++++	+++++	+++++	+++++	+++++	+++++	++++
Knowledge sharing platform	Comprehensive evaluation	*****	****	*****	*****	***	**	**	**	**
	Extent of interaction between participants(strong or weak)	*****	****	***	****	**	**	***	**	**
	The communication atmosphere created(strong or weak)	*****	*****	*****	*****	***	***	***	**	**
Knowledge enhancement function	Teacher-student relationship and degree of communication(strong or weak)	+++++	+++++	+++++	+++++	+++++	+++++	+++++	+++++	++++
	Comprehensive evaluation	*****	*****	*****	*****	++++	++++	++++	++++	++++
	Cohesion education(significant or not)	*****	*****	*****	*****	***	**	0	0	0
Knowledge connection(significant or not)	Knowledge connection(significant or not)	*****	*****	*****	*****	***	**	0	0	0
	Knowledge incubation(significant or not)	*****	*****	*****	*****	***	**	0	0	0
	Comprehensive evaluation	+++++	+++++	+++++	+++++	+++++	+++++	+++++	+++++	++++

Note(s): Remarks: The number of (*) in the table means strong or weak, whether or not, significant or not. The number of (+) means the comprehensive evaluation, equal to the average number of (*)

university. Technical means and knowledge sharing platforms are environmental stimuli for knowledge enhancement. These two forces form the stimulus for the knowledge enhancement of corporate university. The connection between sorting out the knowledge process within the enterprise and sorting out the external knowledge process of enterprises has been verified to be an effective strategic action (Figure 2).

6. Findings of the configural model

6.1 Fuzzy-set qualitative comparative analysis

fsQCA (Ragin, 2008) is used to configure factors that affect knowledge enhancement. In this study, the coded data of references and case studies are taken as the measurement basis of the cause variables and result variables, and then the corresponding material support is found in describing the matching of knowledge management and knowledge enhancement of corporate universities. The method of fuzzy-set qualitative comparative analysis is to score each variable according to the information of sample cases, and then assign values to the variables of the fuzzy set one by one (Douglas et al., 2020). The fsQCA starts with the calibration of data into fuzzy sets. Conditions sufficient and necessary are examined once after calibration is done and dusted. Table 4 shows the necessary condition analysis. As shown in Table 4, each antecedent condition affects the necessity of knowledge enhancement and non-knowledge enhancement that does not exceed 0.9, which does not constitute a necessary condition. This means that the single knowledge management as the antecedent condition has a weaker interpretation of knowledge enhancement. Therefore, the following will incorporate these antecedents into fsQCA to further explore the configuration of high knowledge enhancement and non-knowledge enhancement of corporate universities. fsQCA offers truth tables conferring to the causal conditions that are examined beside with their consistency level. This study examines the intermediate solutions that emerge from the Boolean algorithm to increase the knowledge of adequate conditions that drive towards desired results. The result of the knowledge enhancement presented as an intermediate solution (Ragin, 2008) in Table 5. For the expression of conditional configuration results, this paper refers to the method of Fiss (2011), the presence of cause conditions are represented by “●”, the cause conditions do not present are represented by “⊗”, and a blank to indicate that the cause conditions are irrelevant to the result.

The intermediate solution is obtained by counterfactual analysis, which assuming that the presence of each conditional variable may improve the function of knowledge enhancement of corporate universities. As shown in Table 5, fuzzy-set qualitative comparative analysis shows that there are three configurations (paths) of knowledge management that promote knowledge enhancement function, and the consistency indexes of the three configurations

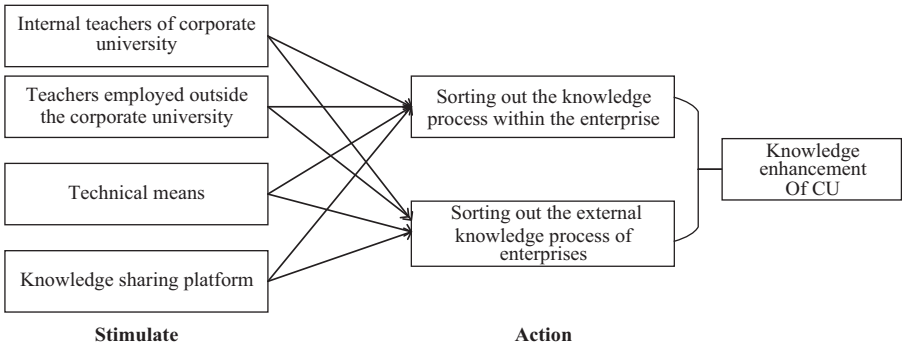


Figure 2.
“Stimulate-action”
model of knowledge
enhancement of
corporate university

Table 4.
Necessity test of knowledge enhancement of corporate universities

Conditional variable	Result variable	
	High knowledge enhancement	Non-knowledge enhancement
IT	0.826456	0.339056
~IT	0.295647	0.793378
TEO	0.398530	0.472103
~TEO	0.700961	0.635806
SKPW	0.884681	0.437155
~SKPW	0.196156	0.650521
SEKP	0.694743	0.541999
~SEKP	0.406444	0.567750
TM	0.517241	0.583078
~TM	0.632561	0.579399
KSP	0.666478	0.429184
~KSP	0.472018	0.721030

Source(s): Calculated through fsqca3.0

Condition variable	High knowledge enhancement (High KE)			Non-knowledge enhancement (Non-KE)	
	H1	H2	H3	NH1	NH2
Internal teachers of corporate university(IT)	●	●	●	⊗	⊗
Teachers employed outside the corporate university(TEO)		●	⊗	⊗	●
Sorting out the knowledge process within the enterprise(SKPW)	●	●	●	⊗	⊗
Sorting out the external knowledge process of enterprises(SEKP)	●	●	⊗		●
Technical means(TM)	⊗		●		●
Knowledge sharing platform(KSP)		●	●	⊗	
raw coverage	0.511023	0.330695	0.305823	0.464746	0.270386
unique coverage	0.150367	0.0384398	0.114189	0.236051	0.0416922
consistency	0.875969	0.881024	0.814759	0.876301	0.832076
Overall solution consistency		0.90169		0.885316	
Overall solution coverage		0.663652		0.506438	

Table 5.
Configuration of knowledge management effecting high and non-knowledge enhancement

Source(s): Calculated through fsqca3.0

are 0.875969, 0.881024 and 0.814759 respectively, which shows that the three configurations are all sufficient conditions for knowledge enhancement. The consistency index of the solution is 0.90169, which further shows that the three configurations covering most of the cases are also sufficient conditions for knowledge enhancement. The coverage of the model solution is 0.663652, indicating that the three configurations explain about 70% of the reasons for knowledge enhancement. At the same time, it is assumed that the absence of each condition variable may lead to non-knowledge enhancement. The fuzzy-set qualitative comparative analysis shows that there are two configurations (paths) of knowledge management that produce non-knowledge enhancement function. These two configurations not only constitute sufficient conditions for non- knowledge enhancement, but also explain the reasons that lead to nearly 60% of non- knowledge enhancement. Table 5 depicts configuration of knowledge management effecting high and non-knowledge enhancement.

In order to better compare the differences of configurations of different knowledge management elements of corporate universities, this paper summarizes the following three configurations (paths) that affect knowledge enhancement, namely according to the results of statistical analysis, namely, the type of strengthening the knowledge process, the type of stronger teacher network, and the type of stronger knowledge ba (Figure 3). At the same time, two kinds of influence configurations (paths) of knowledge management elements that did not produce knowledge enhancement are summarized. The specific explanations for each combination are as follows:

6.2 Configuration analysis of influencing factors of knowledge enhancement

6.2.1 The type of strengthening the knowledge process. $H1: IT \times SKPW \times SEKP \times \sim TM$. The configuration shows that the most important element of the knowledge enhancement of a corporate university is to sort out the internal and external knowledge processes of the enterprise, and at the same time employ internal teachers who are more familiar with enterprise business knowledge and industry knowledge. In this configuration, technical means such as hardware equipment do not play a role and do not affect corporate universities to undertake knowledge aggregation activities.

This path indicates that the process of knowledge management has an interactive effect. With the continuous improvement of corporate university functions, corporate universities will further become the interface of knowledge integration between various departments within the enterprise and between the enterprise and external stakeholders. Corporate universities identify the knowledge weaknesses and required knowledge of business departments through sorting out internal processes, and still need to obtain knowledge and skills that are conducive to knowledge creation through the external network of enterprises, and then transfer the acquired knowledge and skills to the required departments. Through the networking of external contacts with the enterprise, the knowledge of enterprise stakeholders is internalized into the enterprise as much as possible. In this path, the internal and external knowledge process complement each other.

6.2.2 The type of stronger teacher network. $H2: IT \times TEO \times SKPW \times SEKP \times KSP$. This configuration shows that the corporate university pays great attention to the balance of internal and external teachers, and continuously cooperates with internal and external knowledge sources to acquire more knowledge resources, build internal and external teacher networks, and at the same time strengthen the combing of internal and external knowledge processes of the enterprise. It makes the teaching and guidance activities of corporate

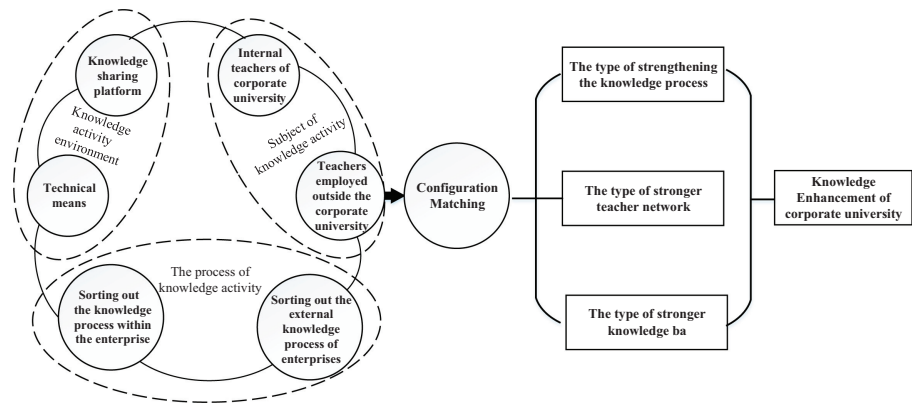


Figure.3.
Configuration model
matching the type of
corporate university
with knowledge
enhancement

universities smoother, stimulates more innovative ideas, and provides a corresponding environment and atmosphere for knowledge creation.

6.2.3 The type of stronger knowledge ba. H3: $IT \times \sim TEO \times SKPW \times \sim SEKP \times TM \times KSP$. Knowledge ba is to promote knowledge exchange and create related shared spaces, including any combination of physical space, virtual space and mental space. This configuration shows that corporate universities can collect and expand external knowledge resources through the improvement of information technology and knowledge base technology to make up for the lack of external teachers and external knowledge combining capabilities. The innovation atmosphere is an important source of enterprise innovation and learning. Students and teachers interact, share and stimulate the knowledge base on the basis of complementary and compatible knowledge, and improve the overall creativity of the team. Corporate universities form a mixed environment of in-depth interaction between teachers and students, promote the accumulation and innovation of team knowledge through learning and discussion, fully share accumulated and sorted knowledge with various business departments, form collective wisdom, and enhance the knowledge ba of corporate universities.

6.3 Configuration analysis of influencing factors of non-knowledge enhancement

In view of the causal asymmetry of the fsQCA method, it means that the emergence of a certain result requires different “cause combinations” to explain separately (Douglas *et al.*, 2020). Therefore, in order to explore the driving mechanism of knowledge function more comprehensively and deeply, this paper further analyzes the configuration (path) that leads to the improvement of non-knowledge enhancement, and the result is shown in Table 5.

6.3.1 Configuration NH1: $\sim ITN \times \sim IKP \times \sim EKP \times \sim IKB$. The configuration shows that once there is a lack of high internal teachers, high internal and external knowledge processes, and no emphasis on building a knowledge sharing platform, the knowledge management activities of corporate universities will become inefficient and the knowledge enhancement function will be weak. The possible reason is that corporate universities are fundamentally different from traditional training centers. The purpose of establishing corporate universities is to promote enterprise reform and integrate the industrial chain, not just the function of talent training (Baporikar, 2015). However, inefficient corporate universities still assume the functions of the original training center, have not contacted with the business department of the enterprise, and ignored the introduction of valuable knowledge and information, which cannot achieve the effect of effective knowledge process sorting. Moreover, corporate universities do not attach importance to the cultivation of internal teachers, and mainly rely on hiring external teachers to teach students. Concerning the professional technology and other problems of the enterprise, external teachers cannot have a thorough understanding, so they cannot solve the practical problems of technicians in professional departments, and the effect of enterprise knowledge training is not good. Another reason for the non-knowledge enhancement is that the corporate university lacks internal teachers and combs the internal knowledge process of the enterprise, which makes it impossible for the trainees to apply what they have learned in combination with the actual work, and cannot carry out workplace learning and build a platform for effective communication and knowledge sharing. Typical examples of this configuration are Taida College, Huali College and Kingdee University.

6.3.2 Configuration NH2: $\sim ITN \times ETN \times \sim IKP \times EKP \times TKB$. This combination shows that the inefficient knowledge management of corporate universities is due to the lack of internal teachers and internal knowledge processes, and only attaches importance to external teachers, external knowledge processes and technical means. Although the corporate university has a fixed place and good basic teaching facilities, it cannot make the corporate university become the source of motivation for organization development and transformation of the enterprise besides undertaking the function of talent training without paying attention to the investment of internal teachers. Inefficient corporate universities mainly rely on

external teachers to teach students, while the external teachers are not clear about the operations of the enterprises affiliated with the corporate universities and the strategic intentions of the top managers, and the knowledge they impart generally lacks practicality and pertinence, which cannot be truly and effectively matched with the actual business of the students. Corporate universities do not have a clear understanding of the knowledge weaknesses of various departments of the enterprise, and all the external knowledge is introduced without evaluation, which makes the corporate university and the company strategy does not match. It cannot transform all kinds of knowledge and form the unique core resources of the enterprise, and even unable to provide comprehensive and more targeted knowledge services for the enterprise.

Comparing the five configurations that affect knowledge enhancement activities, it is found that the knowledge management that affects whether the knowledge enhancement function of corporate university is effective or not is asymmetric, and the paths of non-knowledge enhancement are not the opposites of the paths of high knowledge enhancement activities. That is, the combination of high external knowledge processes and high external teachers, high internal teachers and high knowledge sharing platforms can promote the improvement and development of the knowledge enhancement function of corporate universities (H2). However, when high external knowledge processes and high external teachers, low internal teacher and low internal knowledge processes are combined, the knowledge enhancement function of corporate universities will be inefficient (NH2).

7. Discussion and conclusions

From the multi-case analysis of 34 corporate universities, this study explored the factors and forces influencing knowledge enhancement and detected the relationships among these factors. As shown in the research findings, knowledge enhancement is driven by the subject of knowledge management and the environment of knowledge management, including internal teachers of corporate university, teachers employed outside the corporate university, technical means and knowledge sharing platform. Meanwhile, corporate universities ought to adopt right strategies including sorting out the knowledge process within the enterprise and sorting out the external knowledge process of enterprises. The “stimulate-action” model build up in this study reflects that knowledge enhancement is the combinational result of external and internal corporate university efforts. At the same time, from the perspective of asymmetry, this paper constructs a mechanism for synergistic matching of various elements of knowledge management.

7.1 Theoretical implications

First of all, this paper puts forward an integrated analysis framework for knowledge enhancement of corporate universities. Previous studies only constructed a theoretical model of knowledge management or examined the linear relationship between some factors and knowledge enhancement (Allen, 2010; Scarso *et al.*, 2017; Petr and Elena, 2018), and did not explore the influence mechanism of the configuration effect of the subject of knowledge management, the process of knowledge management, and the environment of knowledge management on knowledge enhancement, and there are few researches on configuration analysis that take knowledge management as an influencing factor. The paper draws lessons from the existing research findings of knowledge management, based on the subject of knowledge management, the environment of knowledge management and the process of knowledge management (SEP), combined with the knowledge management characteristics of Chinese corporate universities, this paper identifies six elements of knowledge management through the case study, including internal teachers of the corporate university, teachers

employed outside the corporate university, sorting out the knowledge process within the enterprise, sorting out the external knowledge process of enterprises, technical means and knowledge sharing platform. At the same time, this paper explores the correlation among influencing factors and constructs a “stimulus-active” model.

Then, it makes up for the lack of integrity of the existing literature on the investigation of factors affecting knowledge enhancement of corporate university. At the same time, this paper refines and extends the knowledge management framework, and further expands the application of the knowledge management framework in explaining the complexity of cause and effect. Previous studies have been limited to a certain level of the subject of knowledge management (Alagaraja and Li, 2015; Tsipes *et al.*, 2016; Slettli, 2017), the process of knowledge management (Martin, 2015; Chen *et al.*, 2019) or the environment of knowledge management (Nonaka *et al.*, 2000), and the mechanism of the synergy of the three types of factors affecting the knowledge management of corporate universities is still unclear. This paper uses the fsQCA method to empirically explore the concurrent synergy effect and linkage matching mode of the multiple conditions of knowledge management in the knowledge enhancement of corporate universities, summarize the combinations of key knowledge management that match the corporate university knowledge enhancement, and finds three paths that affect the function of corporate universities by integrating internal and external teachers, internal and external knowledge processes and knowledge ba, which is helpful to solve the black box of knowledge enhancement of corporate university. Different from most studies use statistical regression to study the marginal net effect of a single condition of the knowledge enhancement factors (Alavi and Leidner, 2001; Dalkir, 2005; Bamel and Bamel, 2018; Denford and Ferriss, 2018), this paper uses fsQCA method to explore the complex interaction among many antecedents from the perspective of configuration, which is suitable for exploring the combination of elements, rather than the influence of a single element on the results (Douglas *et al.*, 2020). As shown in Table 4, the influence of each prerequisite does not exceed 0.9, which means that the single knowledge management as a single condition has a weak explanation for knowledge enhancement and cannot alone constitute a necessary condition for corporate university knowledge enhancement. The complex combination of the six elements of knowledge management can contribute to the knowledge enhancement of corporate universities. As shown in Table 5, none of the six key factors obtained from the grounded analysis are missing in the compound solution of the conditional configuration, which indicates that the element composition in the configuration is diverse and there is a synergistic effect of multiple factors. On the other hand, there is not only one path for the compound solution of the antecedent conditions of each corporate university. Each path is also a combination of 4–5 single factors, which also shows that each path is a combination of key factors. Only by emphasizing the overall synergy of the three factors can corporate universities promote the knowledge enhancement of corporate university.

Moreover, this study uses the fsQCA method to find that the driving mechanism of knowledge management has causal asymmetry. The path that leads to efficient knowledge enhancement and the inefficient path without knowledge enhancement are not the opposite. That is to say, it is not possible to directly derive the combination of non-knowledge enhancement based on the negative side of the combination of knowledge management paths that produce knowledge enhancement. The findings of this study have improved the interpretation of knowledge enhancement of corporate university, which is helpful to explore the phenomenon of knowledge management affected by various factors. There is an interdependent configuration effect among the factors that affect the knowledge enhancement of corporate university. There are an interdependent configuration effects among various factors in the configuration of corporate university knowledge enhancement. The knowledge enhancement of corporate university depends on how an individual knowledge management

factor is interrelated with other factors, and the combination of these factors can play a better role than a single factor. From the perspective of configuration, different combinations of elements have different functions. The elements within the configuration are strong or weak. As long as the structural balance between the factors is formed, the configuration can achieve the knowledge and ability enhancement of the corporate university.

Finally, unlike corporate universities established by enterprises in developed countries, Chinese corporate universities have their unique methods of growth and knowledge management. The knowledge activities of Chinese corporate universities serve the enterprises to which they belong. With the improvement of the knowledge enhancement function of Chinese corporate universities, the role of internal teachers and internal knowledge processes of enterprise have become more prominent. Although the existing research holds that the knowledge enhancement of corporate universities mainly depends on external teachers and external resources (Guerci *et al.*, 2010; Antonelli *et al.*, 2013; Cappiello and Pedrini, 2017). However, this paper finds that no matter which Chinese corporate university has more than one way to promote knowledge management, but internal teachers and sorting out the knowledge process within the enterprise are necessary conditions of the excellent sample corporate universities. Taking 34 corporate universities as examples, the method of fuzzy-set qualitative comparative analysis holds that the conditions that appear in both parsimonious solutions and intermediate solutions are the core elements, indicating that they are the condition that have an important influence on the results (Fiss, 2011; Misangyi *et al.*, 2017). By comparing the three configurations of knowledge enhancement of corporate university, it is found that the parsimonious solution of internal teachers of the corporate university and sorting out the knowledge process within the enterprise also appear in the intermediate solution, which shows that it plays a core role in improving the knowledge enhancement function of the corporate university. At the same time, high internal teachers of the corporate university and high sorting out the knowledge process within the enterprise have complementary effects. That is, if the internal teachers and sorting out the internal knowledge process of the enterprise are strong, as long as the single condition of efficiently sorting out the external knowledge process is satisfied, or both the two conditions of technical means and building a knowledge-sharing platform are met at the same time, the more effective knowledge enhancement function of the corporate university can be promoted. For Chinese corporate universities, the management of external knowledge resources is not the core reason for the knowledge enhancement of corporate university. This also provides a new extension for the organizational rationality and application of the corporate university wheel model constructed by Prince and Stewart (2002) in the Chinese context.

7.2 Practice implications

First of all, this paper selects a more effective combination path of knowledge management for the knowledge enhancement of corporate universities. Our research guides corporate universities to pay attention to the coordination mechanism and integrity of knowledge management. The subject of knowledge activity, the process of knowledge activity and knowledge activity environment are all indispensable in the configuration that affects the knowledge enhancement of corporate universities.

Secondly, this paper provides a practical reference for corporate universities to effectively cultivate knowledge management capabilities. The purpose of enterprise establishing corporate universities is to promote enterprise reform and integrate industrial chains, not just talent training. However, the inefficient corporate universities still assume the functions of the original training center, does not contact with the business department of the enterprise, and ignores the introduction of valuable knowledge and information, thus failing to achieve the effect of effective knowledge process sorting. This paper finds that the excellent case corporate

universities pay more attention to the ability of sorting out the knowledge process within the enterprise, rather than just introducing external resources. In terms of the introduction of external resources, enterprise universities should focus more on industry-based curriculum and revise it periodically to internalize external knowledge. Corporate universities with strong knowledge management capabilities pay more attention to collecting the knowledge needs of various business departments, and explore the knowledge weaknesses of the business departments by combing business processes. Such corporate universities also enable various departments to strengthen exchanges, cooperation and knowledge linkage through training, building case sharing platforms, and organizing seminars. The foundation of enterprise development is the sustainable and sound development of business. The primary task of corporate universities is to serve enterprises and cannot exist in isolation from enterprises. Inefficient corporate universities have not considered the strategy and business development of enterprises at all in the development process, which is obviously contrary to the foundation of corporate universities. Therefore, this paper finds that the case corporate universities with high knowledge management capabilities focus on sorting out the internal knowledge process within the enterprise, which closely links the enterprise strategy and business. Only in this way can corporate universities further become the knowledge center of the enterprise. These have some references for other developing corporate universities.

At last, at present, many corporate universities with weak knowledge management capabilities mainly employ external professionals, such as university professors, scholars, consultants, and vocational trainers as teachers. Most of them have the most cutting-edge academic ideas and concepts, broad vision and rich experience in enterprise training, but they generally lack practicality and pertinence, and cannot be truly and effectively integrated with the actual situation of enterprises and the working conditions of employees. In addition, the teachers and courses of corporate universities with low knowledge management capabilities are mainly imported from outside, making how to integrate foreign courses and teachers with the unique culture of the enterprises has become an urgent problem for corporate universities. However, corporate universities with strong knowledge management capabilities focus on cultivating teachers in enterprises. The internal teachers of the enterprise are familiar with the actual situation of the industry and the enterprise, and have personal experience of the problems in the operation and development of the enterprise. More importantly, they do not have the problem of cultural integration with the enterprise, and can interpret training needs from the perspective of insiders, and accurately analyze and judge the problems existing in the enterprise. Therefore, according to the experience of the corporate universities with strong knowledge management capabilities among the 34 corporate universities in this paper, corporate universities should focus on tapping the potential of internal managers of enterprises, and allow the managers of enterprises to work part-time as internal teachers of enterprises, which can effectively make up for the deficiencies and shortcomings of external professional training, and truly form a training mechanism of corporate universities with complementary advantages of internal teachers and external professionals. In a few corporate universities with strong knowledge management capabilities, the teachers who teach include most of the board members and middle and senior managers of enterprises, and they will frequently participate in the teaching of the courses. The chairman of enterprises affiliated with corporate universities usually also participates in the lectures. Therefore, how to train managers into enterprise coaches is a key task faced by corporate universities. This is not only a successful experience followed by corporate universities with strong knowledge management capabilities, but also the new trend nowadays.

8. Limitations and future research

The antecedent configuration proposed in this paper for knowledge management to promote the knowledge enhancement of corporate universities still has the following limitations: Due

to the limitation of the degree of detail and quantity of cases, the number of antecedents proposed in this study is limited to a certain extent. Besides, due to the scarcity of literature research on the failure of knowledge enhancement of corporate universities, the reasons for non-knowledge enhancement of corporate universities have not been explored in-depth, which can be further discussed in the future.

References

- Abel, A.L. and Li, J. (2012), "Exploring the corporate university phenomenon: development and implementation of a comprehensive survey", *Human Resource Development Quarterly*, Vol. 23 No. 1, pp. 103-128.
- Akram, A. (2002), "The corporate university model for continuous learning, training and development", *Education and Training*, Vol. 44 No. 2, pp. 76-81.
- Alagaraja, M. and Li, J. (2015), "Utilizing institutional perspectives to investigate the emergence, rise, and (relative) decline of corporate universities", *Human Resource Development International*, Vol. 18 No. 1, pp. 4-23.
- Alavi, M. and Leidner, D.E. (2001), "Review: knowledge management and knowledge management systems: conceptual foundations and research issues", *MIS Quarterly*, Vol. 25 No. 1, pp. 107-136.
- Allen, M. (2010), "Corporate universities 2010: globalization and greater sophistication", *Journal of International Management Studies*, Vol. 5 No. 1, pp. 48-53.
- Andresen, M. and Lichtenberger, B. (2007), "The corporate university landscape in Germany", *Journal of Workplace Learning*, Vol. 19 No. 2, pp. 109-123.
- Antonelli, G., Cappiello, G. and Pedrini, G. (2013), "The corporate university in the European utility industries", *Utilities Policy*, Vol. 25 No. 4, pp. 33-41.
- Bamel, U.K. and Bamel, N. (2018), "Organizational resources, KM process capability and strategic flexibility: a dynamic resource-capability perspective", *Journal of Knowledge Management*, Vol. 22 No. 7, pp. 1555-1572.
- Baporikar, N. (2015), "Role of corporate universities in higher education", *International Journal of Applied Management Sciences and Engineering*, Vol. 2 No. 1, pp. 30-43.
- Bhatt, G. (2001), "Knowledge management in organizations: examining the interaction between technologies, techniques, and people", *Journal of Knowledge Management*, Vol. 5 No. 1, pp. 68-75.
- Brunswick, S. and Vanhaverbeke, W. (2014), "Open innovation in small and medium-sized enterprises: external knowledge sourcing strategies and internal organizational facilitators", *Journal of Small Business Management*, Vol. 53 No. 4, pp. 1241-1263.
- Cappiello, G. and Pedrini, G. (2017), "The performance evaluation of corporate universities", *Tertiary Education and Management*, Vol. 23 No. 3, pp. 304-317.
- Cerchione, R. and Esposito, E. (2017), "Using knowledge management systems: a taxonomy of SME strategies", *International Journal of Information Management*, Vol. 37 No. 1, pp. 1551-1562.
- Chen, Y.Q., Xu, Y.S. and Zhai, Q.G. (2019), "The knowledge management functions of corporate university and their evolution: case studies of two Chinese corporate universities", *Journal of Knowledge Management*, Vol. 23 No. 10, pp. 2086-2112.
- Clinton, M.S., Merritt, K.L. and Murray, S.R. (2009), "Using corporate universities to facilitate knowledge transfer and achieve competitive advantage: an exploratory model based on media richness and type of knowledge to be transferred", *International Journal of Knowledge Management*, Vol. 5 No. 4, pp. 43-59.
- Dalkir, K. (2005), *Knowledge Management in Theory and Practice*, Elsevier Butterworth-Heinemann, Oxford.
- Darroch, J. (2005), "Knowledge management, innovation and firm performance", *Journal of Knowledge Management*, Vol. 9 No. 3, pp. 101-115.

- Dealtry, R. (2006), "The corporate university's role in managing an epoch in learning organization innovation", *Journal of Workplace Learning*, Vol. 18 No. 5, pp. 313-320.
- Dealtry, R. and Rademakers, M. (2005), "Corporate universities: driving force of knowledge innovation", *Journal of Workplace Learning*, Vol. 17 No. 2, pp. 130-136.
- Denford, J.S. and Ferriss, A. (2018), "Absorption, combination and desorption: knowledge-oriented boundary spanning capacities", *Journal of Knowledge Management*, Vol. 22 No. 7, pp. 1425-1441.
- Doody, K.J. (2001), "Implementing a corporate-university relations program", *Research-Technology Management*, Vol. 44 No. 5, pp. 42-61.
- Douglas, E.J., Shepherd, D.A. and Prentice, C. (2020), "Using fuzzy-set qualitative comparative analysis for a finer-grained understanding of entrepreneurship", *Journal of Business Venturing*, Vol. 35 No. 1, pp. 59-90.
- Duysters, G. and Lokshin, B. (2011), "Determinants of alliance portfolio complexity and its effect on innovative performance of companies", *Journal of Product Innovation Management*, Vol. 28 No. 4, pp. 570-585.
- Eisenhardt, K.M. and Graebner, M.E. (2007), "Theory building from cases: opportunities and challenges", *Academy of Management Journal*, Vol. 50 No. 1, pp. 25-32.
- Elena, A., Melewar, T.C. and Pantea, F. (2018), "Examining the influence of corporate website favorability on corporate image and corporate reputation: findings from fsQCA", *Journal of Business Research*, Vol. 36 No. 1, pp. 1-18.
- Fiss, P.C. (2007), "A set-theoretic approach to organizational configurations", *Academy of Management Review*, Vol. 32 No. 4, pp. 1180-1198.
- Fiss, P.C. (2011), "Building better causal theories: a fuzzy set approach to typologies in organization research", *Academy of Management Journal*, Vol. 54 No. 2, pp. 393-420.
- Gao, F., Li, M. and Clarke, S. (2008), "Knowledge, management and knowledge management in business operations", *Journal of Knowledge Management*, Vol. 12 No. 2, pp. 3-17.
- Gold, A., Malhotra, A. and Segard, A. (2001), "Knowledge management: an organizational capabilities perspective", *Journal of Management Information Systems*, Vol. 18 No. 1, pp. 185-214.
- Guerci, M., Bartezzaghi, E. and Solari, L. (2010), "Training evaluation in Italian corporate universities: a stakeholderbased analysis", *International Journal of Training and Development*, Vol. 14 No. 4, pp. 291-308.
- Khuram, S., Sami, U.B., Ahmed, F.I. and Farhan, A. (2016), "Integrating knowledge management strategies and processes to enhance organizational creativity and performance: an empirical investigation", *Journal of Modelling in Management*, Vol. 11 No. 1, pp. 154-179.
- Martin, C. (2015), "Knowledge management and innovation in knowledge-based and high-tech industrial markets: the role of openness and absorptive capacity", *Industrial Marketing Management*, Vol. 47 No. 11, pp. 143-146.
- Meister, J.C. (1997), "Market audits of corporate universities", *Corporate University Review*, Vol. 5 No. 1, pp. 26-34.
- Meister, J. (2001), "The brave new world of corporate education", *The Chronicle of Higher Education*, Vol. 9 No. 2, pp. 10-13.
- Miles, M.B. and Huberman, A.M. (1994), *Qualitative Data Analysis: An Expanded Sourcebook*, 2nd ed., Sage, Thousand Oaks, CA.
- Misangyi, V.F. and Acharya, A.G. (2014), "Substitutes or complements? A configural examination of corporate governance mechanisms", *Academy of Management Journal*, Vol. 57 No. 6, pp. 1681-1705.
- Misangyi, V.F., Greckhamer, T., Furnari, S. and Fiss, P.C. (2017), "Embracing causal complexity: the emergence of a neo-configurational perspective", *Journal of Management*, Vol. 43 No. 1, pp. 255-282.

- Mors, M.L. (2010), "Innovation in a global consulting firm: when the problem is too much diversity", *Strategic Management Journal*, Vol. 31 No. 8, pp. 841-872.
- Nixon, J. and Helms, M. (2002), "Corporate universities vs higher education institutions", *Industrial and Commercial Training*, Vol. 34 No. 4, pp. 144-151.
- Nonaka, I., Toyama, R. and Konno, N. (2000), "SECI, ba and leadership: a unified model of dynamic knowledge creation", *Long Range Planning*, Vol. 33 No. 1, pp. 5-34.
- Obermayer, N. and Toth, V.E. (2019), "Organizational dynamics: exploring the factors affecting knowledge sharing behavior", *Kybernetes*, Vol. 49 No. 1, pp. 165-181.
- Paton, R., Peters, G. and Storey, J. (2005), "Corporate universities as strategic learning initiatives", *The Handbook of Corporate University Development: Managing Strategic Learning Initiatives in the Public and Private Domains*, Gower, London.
- Patrucco, A.S., Pellizzoni, E. and Buganza, T. (2017), "The design process of corporate universities: a stakeholder approach", *Journal of Workplace Learning*, Vol. 29 No. 4, pp. 304-318.
- Petr, P. and Elena, A.S. (2018), "With or without CU a comparative study of efficiency of European and Russian corporate universities", *Journal of Intellectual Capital*, Vol. 19 No. 1, pp. 96-111.
- Prince, C. and Allison, S. (2003), "Corporate university developments in the UK SME sector", *Journal of Workplace Learning*, Vol. 15 No. 5, pp. 229-239.
- Prince, C. and Stewart, J. (2002), "Corporate universities-an analytical framework", *The Journal of Management Development*, Vol. 21 No. 10, pp. 794-811.
- Prof, M.D.G., Vincenzo, M., P. and Jiménez, J. (2014), "Knowledge management practices for innovation: a multinational corporation's perspective", *Journal of Knowledge Management*, Vol. 18 No. 5, pp. 905-918.
- Ragin, C.C. (2008), *Redesigning Social Inquiry: Fuzzy Sets and beyond*, Chicago University Press, Chicago.
- Rheame, L. and Gardoni, M. (2016), "Strategy-making for innovation management and the development of corporate universities", *International Journal on Interactive Design and Manufacturing*, Vol. 10 No. 1, pp. 73-84.
- Sandelands, E. (1997), "Developing a robust model of the virtual corporate university", *Journal of Knowledge Management*, Vol. 1 No. 3, pp. 181-188.
- Scarso, E., Lee, W. and Bolisani, E. (2017), "Corporate universities as knowledge management tools", *VINE Journal of Information and Knowledge Management Systems*, Vol. 47 No. 4, pp. 538-554.
- Schneckenberg, D., Truong, Y. and Mazloomi, H. (2015), "Micro foundations of innovative capabilities: the leverage of collaborative technologies on organizational learning and knowledge management in a multinational corporation", *Technological Forecasting and Social Change*, Vol. 100 No. 5, pp. 356-368.
- Schultz, D. (2015), "The rise and coming demise of the corporate university", *Human Resource Development International*, Vol. 101 No. 5, pp. 62-79.
- Sharma, P. and Pandher, J.S. (2018), "Teachers' professional development through teachers' professional activities", *Journal of Workplace Learning*, Vol. 30 No. 8, pp. 613-625.
- Slettl, K.V. (2017), "Coordinating flows of organizational knowledge: the role of the corporate university", *Russian Management Journal*, Vol. 15 No. 1, pp. 51-68.
- Tsipis, G., Echkalova, N., Sharova, E. and Tovb, A. (2016), "Corporate University as a driver of project culture and competence development", *Procedia - Social and Behavioral Sciences*, Vol. 22 No. 6, pp. 335-342.
- Yang, B., Wang, L. and Mohammed, B.O. (2019), "Improving the organizational knowledge sharing through online social networks: the mediating role of employee motivation", *Kybernetes*, Vol. 49 No. 11, pp. 2615-2632.

Yin, R.K. (2003), *Case Study Research: Design and Methods*, 3rd ed., Sage, London.

Further reading

Rheaume, L. and Gardoni, M. (2015), "The challenges facing corporate universities in dealing with open innovation", *Journal of Workplace Learning*, Vol. 27 No. 4, pp. 112-116.

Shaw, S. (2010), "The corporate university: global or local phenomenon?", *Journal of European Industrial Training*, Vol. 29 No. 1, pp. 21-39.

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